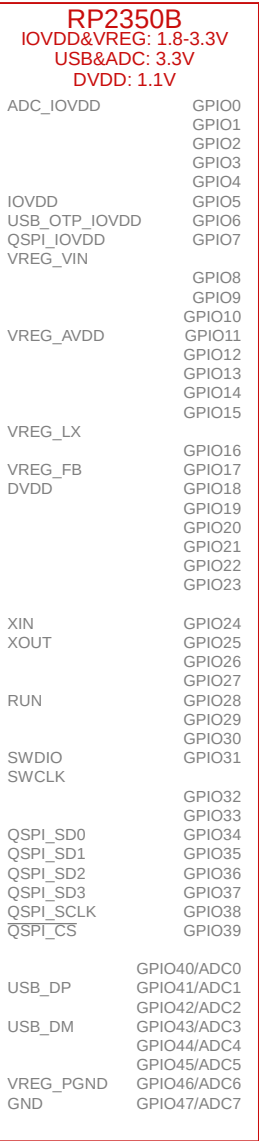


# Pimoroni Pico LiPo 2 (PIM775)

RP2350B MCU

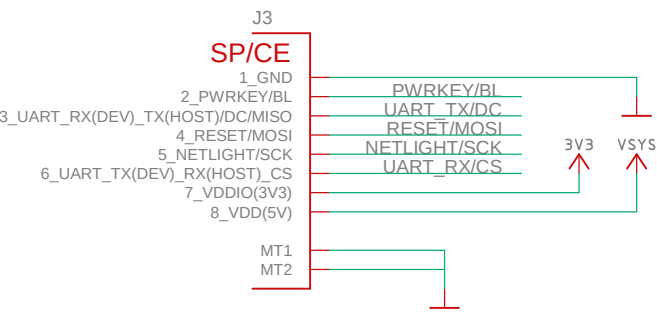
MCU2



|             |    |                         |
|-------------|----|-------------------------|
| GPIO0       | 77 | GP0                     |
| GPIO1       | 78 | GP1                     |
| GPIO2       | 79 | GP2                     |
| GPIO3       | 80 | GP3                     |
| GPIO4       | 1  | GP4_SDA                 |
| GPIO5       | 2  | GP5_SCL                 |
| GPIO6       | 3  | GP6                     |
| GPIO7       | 4  | GP7                     |
| GPIO8       | 6  | GP8                     |
| GPIO9       | 7  | GP9                     |
| GPIO10      | 8  | GP10                    |
| GPIO11      | 9  | GP11                    |
| GPIO12      | 11 | GP12                    |
| GPIO13      | 12 | GP13                    |
| GPIO14      | 13 | GP14                    |
| GPIO15      | 14 | GP15                    |
| GPIO16      | 16 | GP16                    |
| GPIO17      | 17 | GP17                    |
| GPIO18      | 18 | GP18                    |
| GPIO19      | 19 | GP19                    |
| GPIO20      | 20 | GP20                    |
| GPIO21      | 21 | GP21                    |
| GPIO22      | 22 | GP22                    |
| GPIO23      | 23 |                         |
| GPIO24      | 25 | VBUS_SENSE              |
| GPIO25      | 26 | LED_GP25                |
| GPIO26      | 27 | GP26                    |
| GPIO27      | 28 | GP27                    |
| GPIO28      | 36 | GP28                    |
| GPIO29      | 37 |                         |
| GPIO30      | 38 |                         |
| GPIO31      | 39 |                         |
| GPIO32      | 40 | UART_TX/DC              |
| GPIO33      | 42 | UART_RX/CS              |
| GPIO34      | 43 | NETLIGHT/SCK            |
| GPIO35      | 44 | RESET/MOSI              |
| GPIO36      | 45 | PWRKEY/BI               |
| GPIO37      | 46 |                         |
| GPIO38      | 47 |                         |
| GPIO39      | 48 |                         |
| GPIO40/ADC0 | 49 | 1 RN2-1 8 GP26          |
| GPIO41/ADC1 | 52 | 1k 2 RN2-2 7 GP27       |
| GPIO42/ADC2 | 53 | 1k 3 RN2-3 6 GP28       |
| GPIO43/ADC3 | 54 | 1k 4 RN2-4 5 VBAT_SENSE |
| GPIO44/ADC4 | 55 |                         |
| GPIO45/ADC5 | 56 | USER_SW                 |
| GPIO46/ADC6 | 57 | PSRAM_CS                |
| GPIO47/ADC7 | 58 |                         |

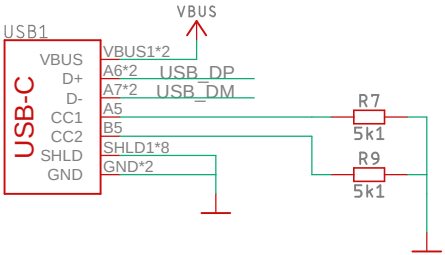


SP/CE Connector

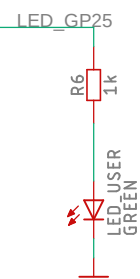


# Pimoroni Pico LiPo 2 (PIM775)

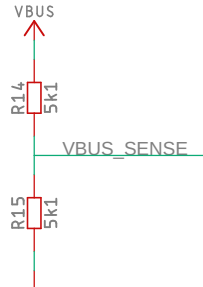
USB-C connector



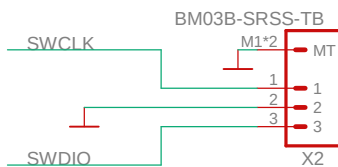
User LED



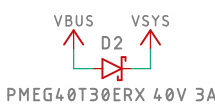
Vbus sense voltage divisor



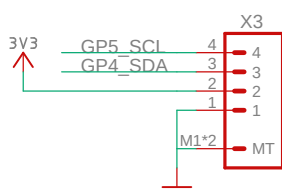
Debug connector



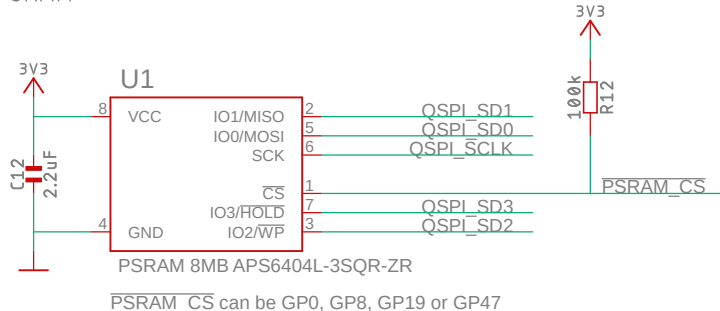
Vbus to Vsys diode connection



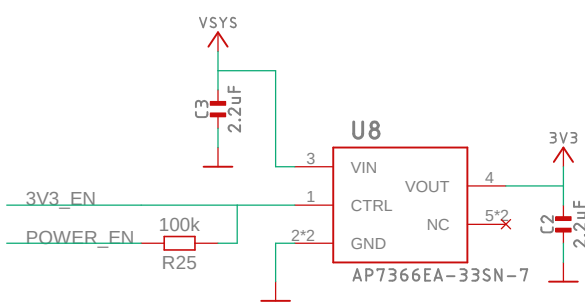
QW/ST connector



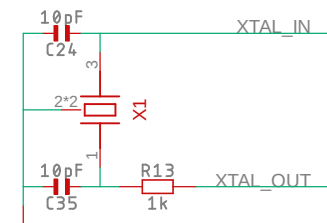
PSRAM



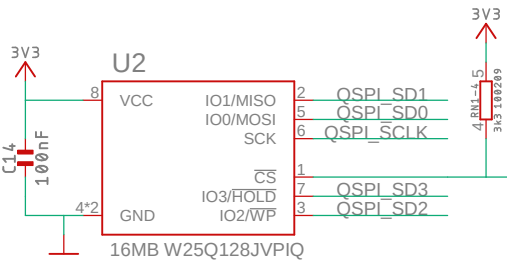
LDO for 3v3 output, low Iq



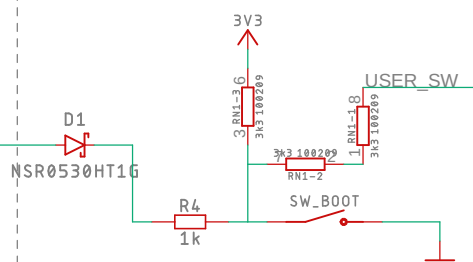
Crystal oscillator



Flash memory

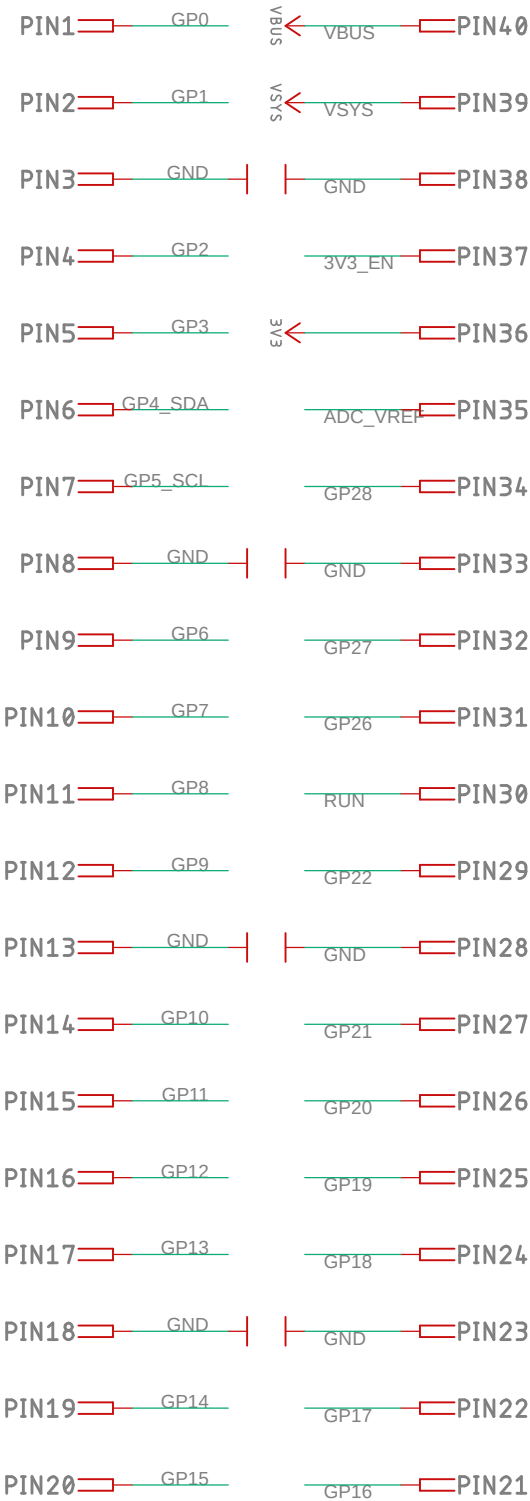


Combined USB boot & user switch



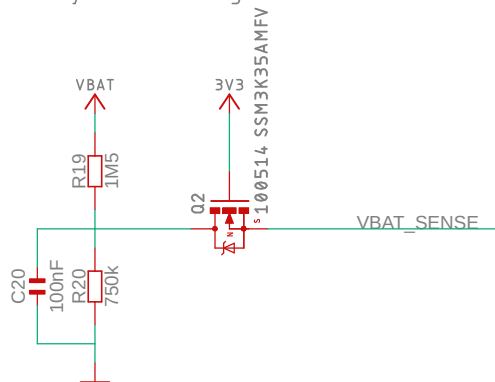
Pimoroni Pico LiPo 2 (PIM775)

Output headers



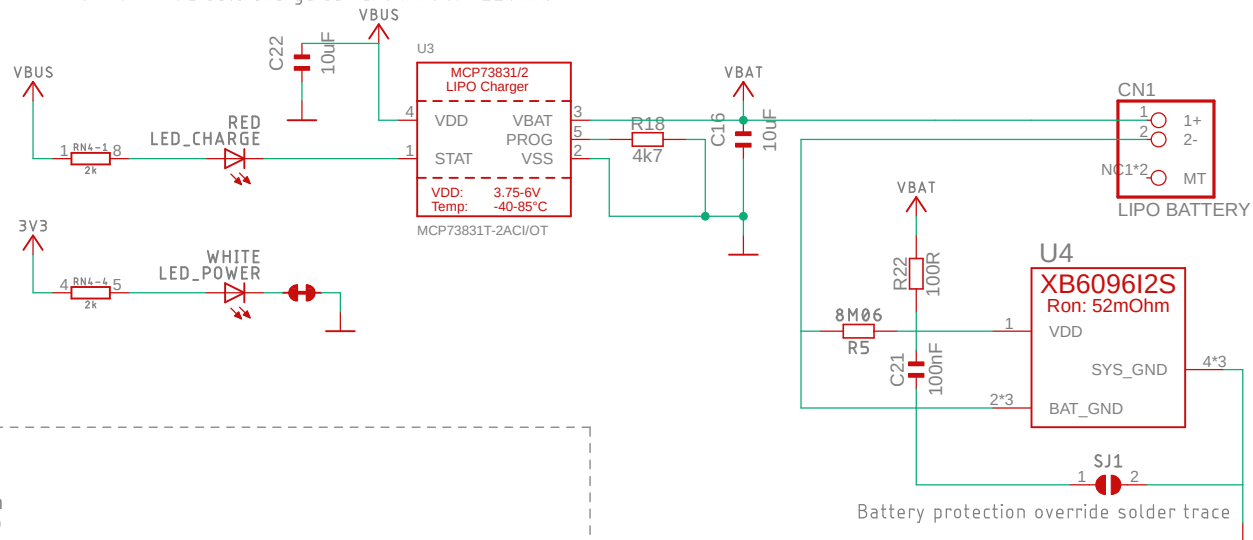
## Pimoroni Pico LiPo 2 (PIM775)

Battery sense voltage divisor



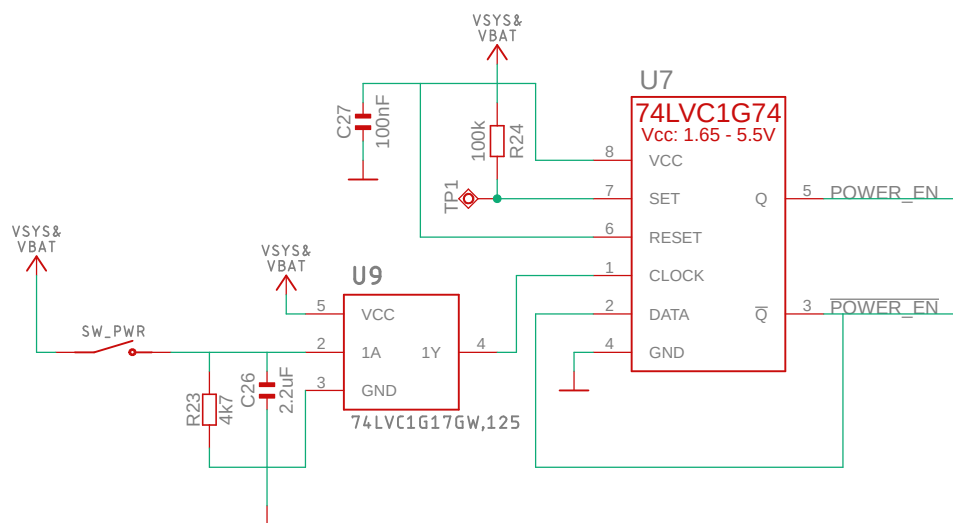
## Lipo charger, protection and connector

4.7k on PROG sets charge current limit to ~220mA.

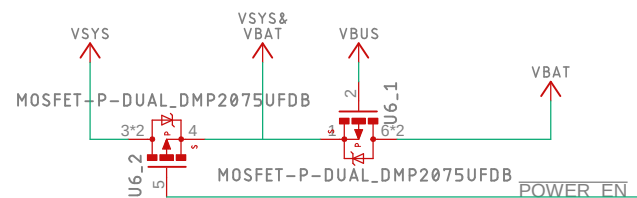


## Power switch toggle circuit

R24 and the PCB capacitor slow down the SET rising slope enough compared to RESET that the chip reliably defaults to on at powerup



## Vbat charge protection and power switch FET switches



Typical discharge power path resistance:  
XB6096I2S: 52mOhm  
DMP2075UFDB: 64mOhm

->

52 + 2\*64 = 180mOhm  
Voltage drop at 1A = 180mV